Scheme and Answer Key for **25901918**

MAHATMA GANDHI UNIVERSITY, KOTTAYAM MGU-UGP (Honours)/MGU-BBA (Honours)/MGU-BCA (Honours) Special Examinations, July 2026 (for 2024 Admission Regular Examinations, October 2025)

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCPHY201

ESSENTIAL MATHEMATICS FOR PHYSICS

(2024 ADMISSION ONWARDS)

Duration: 2 Hours Maximum Marks: 70

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions
Answer any 10 questions
Each question carries 3 marks

1	It measures how much one vector aligns with the direction of another vector OR Projection of one vector onto the direction of the other. Eg. Work W=**F.r,** potential energy of a magnetic dipole -**m.B**	Physical significance- 2 mark; OR explanation using vectors Any one relevant example- 1 mark If the definition of dot product is mentioned- 1 mark
2	A measure of how much two vectors point in the same direction, <**x**, **y**> =||x|| ||y|| cos θ.	Geometrical meaning- 3 mark Equation only- 2 mark
3	When the three vectors are coplanar, any two vectors are parallel or the same, or at least one vector is a zero vector	Any two – 1.5 marks each
4	Using the characteristic equation |**A**-λ**I**|=0, finding eigen values- λ=0,1,2 Finding an eigen vector $(0\ k\ -k)^T$, $(0\ 0\ 0)^T$, $(0\ k\ k)^T$ [This question should come under part B as the procedure is lengthy to some extent]	Characteristic equation -1 mark. Finding an eigen value- 1 mark Finding an eigen vector-1 mark
5	The diagonal elements of a Hermitian matrix are real because each diagonal entry must equal its own complex conjugate. A complex number equals its own conjugate only when its imaginary part is zero For a matrix A to be Hermitian $A=A^\dagger=(A^*)^T$	For the definition of a Hermitian matrix only- 1.5 marks
6	A similarity transformation is a geometric operation or a combination of operations that	Transformation equation only- 1 mark

	changes the size of a figure while preserving its shape, specifically consisting of **dilations** combined with **rigid motions** like rotations, reflections, and translations. Two figures are considered similar if and only if there is a similarity transformation that maps one figure onto the other OR A transformation that can be particularly useful in the conversion of matrices into convenient forms for computation. $A'=S^{-1}AS$	
7	In a general curvilinear coordinate system, the scale factor $h_i = \|\partial \mathbf{r}/\partial u_i\|$	Equation-1.5 marks, Explanation of terms- 1.5 marks. If the scale factors of the Cartesian coordinate system are found- 2 marks
8	Time derivative $\boldsymbol{v} = \frac{d\boldsymbol{r}}{dt} = \frac{dx}{dt}\boldsymbol{i} + \frac{dy}{dt}\boldsymbol{j} + \frac{dz}{dt}\boldsymbol{k}$ OR $\dot{\boldsymbol{r}} = \dot{x}\boldsymbol{i} + \dot{y}\boldsymbol{j} + \dot{z}\boldsymbol{k}$ Physical meaning- velocity	Expression for time derivative- 1.5 marks Physical meaning- 1.5 marks
9	$\nabla = \frac{\partial}{\partial x}\boldsymbol{i} + \frac{\partial}{\partial y}\boldsymbol{j} + \frac{\partial}{\partial z}\boldsymbol{k}$ - In Cartesian coordinate system $\mathbf{A} = A_x\mathbf{i} + A_y\mathbf{j} + A_z\mathbf{k}$ $\nabla \times \boldsymbol{A} = \begin{vmatrix} \boldsymbol{i} & \boldsymbol{j} & \boldsymbol{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$ The proof can be obtained using these relations	Definition of ∇ only- 1 mark; Definition of A – ½ mark Finding $\nabla \times \boldsymbol{A}$- 1.5 marks, Using these relations and obtaining the result- 3 marks
10	Force $\boldsymbol{F} = -\nabla\varphi = (y^3 - 3x^2y)\mathbf{i} + (3xy^2 - x^3 - 2)\boldsymbol{j}$ Work $\int_{(1,1)}^{(2,-1)} \boldsymbol{F}.d\boldsymbol{r} = 2[xy^3 - x^3y - y]_{(1,1)}^{(2,-1)} =$ 18 units	Force- 1.5 marks Work=1.5 marks
11	The total outward flux of a vector field through a closed surface equals the total volume integral of the field's divergence throughout the region enclosed by that surface.	Statement OR equation with terms explained- 3 marks
12	$\iiint_V f(x,y,z)dV$	Correct expression- 3marks

[3x10 = 30]

Part B

Short Essay Type Questions

Answer any 8 questions

Each question carries 5 marks

13	$\|\|\boldsymbol{a}+\boldsymbol{b}\|\| \leq \|\|\boldsymbol{a}\|\| + \|\|\boldsymbol{b}\|\|$ $\boldsymbol{a} = 3\boldsymbol{i} - \boldsymbol{j} + 2\boldsymbol{k}; \boldsymbol{b} = -\boldsymbol{i} + 4\boldsymbol{j}$ $\boldsymbol{a}+\boldsymbol{b} = 2\boldsymbol{i} + 3\boldsymbol{j} + 2\boldsymbol{k}$ $\|\|\boldsymbol{a}+\boldsymbol{b}\|\| = \sqrt{4+9+4} = \sqrt{17}$ $\|\|\boldsymbol{a}\|\| = \sqrt{9+1+4} = \sqrt{14}$ $\|\|\boldsymbol{b}\|\| = \sqrt{1+16} = \sqrt{17}$	Statement- 1 mark Equations for **a** and **b-** 1 mark $\boldsymbol{a}+\boldsymbol{b}$ **:-** 1 mark Norm of $\boldsymbol{a}+\boldsymbol{b}$ **–** 1 mark Norm of $\boldsymbol{a}$- 1 mark Norm of **b** – 1 mark Hence proof- 1 mark All-5 marks
14	Two vectors are orthogonal if <**a**|**b**>=0 Orthonormal- mutually orthogonal and each has unit norm $\boldsymbol{a} = \boldsymbol{j}$ $\boldsymbol{b} = \boldsymbol{i} + \boldsymbol{k}$ **a.b**=0; Hence both vectors are orthogonal	1 mark 1 mark 1 mark 1 mark 1 mark
15	$[A:B] = \begin{bmatrix} 1 & -1 & :1 \\ 1 & 1 & :5 \end{bmatrix}$ Solving to get x= 3 and y= 2	Augmented matrix- 1 mark Transformations- 3 marks Result- 1 mark
16	$\sigma_x\sigma_y = \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix}$; $\sigma_y\sigma_y = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ There is a mistake in the question it should be $\sigma_x\sigma_y$-$\sigma_y\sigma_x$	2.5 marks 2.5 marks Marks should be given if the above two operations were made.
17	For a unitary matrix $A^\dagger A=I$ $A^\dagger = \frac{1}{3}\begin{bmatrix} 2+i & 2 \\ 2 & 2-i \end{bmatrix}$ Finding $A^\dagger A$ and showing it is not equal to I and hence the given Matrix is not unitary |A|=1/3	1 mark 1 mark 2 marks 1 mark
18	$\frac{\partial \phi}{\partial r} = 2rsin\theta cos\emptyset$ $\frac{\partial \emptyset}{\partial \theta} = r^2 cos\theta cos\emptyset$ $\frac{\partial \emptyset}{\partial \emptyset} = -r^2 sin\theta sin\emptyset$ Substitution and final result$\nabla\emptyset = 2rsin\theta cos\emptyset\hat{r} + rcos\theta cos\emptyset\hat{\theta} - rsin\emptyset\hat{\emptyset}$	1 mark 1 mark 1 mark 2 marks
19	$\phi\psi = 2x^2y^2z^3$ $\nabla(\phi\psi) = 4xy^2z^3\boldsymbol{i} + 4x^2yz^3\boldsymbol{j} + 6(xyz)^2\boldsymbol{k}$ $\boldsymbol{\nabla\emptyset} = 2xy\boldsymbol{i} + x^2\boldsymbol{j}$ $\boldsymbol{\nabla\psi} = 2z^3\boldsymbol{j} + 6yz^2\boldsymbol{k}$ Substitution for RHS and obtaining the result	1 mark 1 mark 1 mark 1 mark 1 mark
20	Finding $\nabla . A = 0$	4 marks

	A is Solenoidal OR vector field A has **no net sources or sinks**, meaning the total flow entering any enclosed region exactly equals the total flow leaving it.	Physical meaning- 1 mark
21	In Spherical polar coordinates $dA = r^2 \sin\theta \ d\theta \ d\varphi$ $$A = \int_0^{2\pi} \int_0^{\pi} r^2 \sin\theta \, d\theta \, d\varphi$$ calculating to obtain $$A = r^2 \cdot (2\pi) \cdot (2) = 4\pi r^2$$	2 marks Substitution of limits- 1 mark Obtaining final result- 2 marks
22	W=$\int_0^2 F.dx = \int_0^2 (3x+4)dx = \left[\frac{3x^2}{2} + 4x\right]_0^2 = 14J$	Answer with unit 5 marks